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How would you evaluate the performance of your solution in: POS tagging, NER, dependency parsing?

The accuracy of the POS tagging was fairly accurate. For the abstracts, it performed very well. It struggled some with the GCMD Science Keywords, namely with symbols and scientific notation (such as variables).

The NER output seemed to have multiple issues. For example, it never classified *Venus* as a location. Most of the time, it was classified as an organization, and once a person. Similar to the POS tagging results, the dependency parsing did well with the abstracts, but struggled with symbols in the GCMD keywords.

Overall, the outputs were adequate with standard English, as can be seen with the abstract, but could not handle symbols and certain domain-specific terminology.

Example input and output from: POS tagging, NER, dependency parsing.

Input:

Venus is the ultimate control case for understanding how Earth developed and maintained conditions suited to life.

POS:

('Venus', 'NNP')
('is', 'VBZ')
('the', 'DT')
('ultimate', 'JJ')
('control', 'NN')
('case', 'NN')
('for', 'IN')
('understanding', 'VBG')
('how', 'WRB')
('Earth', 'NNP')
('developed', 'VBD')
('and', 'CC')

('maintained', 'VBD')
('conditions', 'NNS')
)('suited', 'VBN')
('to', 'TO')
('life', 'NN')
('.', '.')

NER:

Venus ORG
Earth LOC

Dependency Parsing:

Venus (nsubj): is
is (ROOT): is
the (det): case
ultimate (amod): case
control (compound): case
case (attr): is
for (prep): case
understanding (pcomp): for
how (advmod): maintained
Earth (nsubj): developed
developed (ccomp): understanding
and (cc): developed
maintained (conj): developed
conditions (dobj): maintained
suited (acl): conditions
to (prep): suited
life (pobj): to
. (punct): is

Explain how the length and quality of your document would affect your lab results.

Quality seemed very important for the results. There are some places in the corpus where the text is convoluted (potentially due to the conversion from equation symbols to plain text), and this caused the output quality to decrease.

Sentence length also seemed to have an effect on the quality of the results. Shorter sentences appeared to be more accurate than longer sentences, but perhaps this is due to human error when interpreting the results.